



01 Introduction

01.01 Introduction concepts, grids, cloud, virtualization and networks classification



What is a computer network

- 2 or more interconnected computing devices connected by a single “technology” are a network
- A distributed system may be confused with a CN but they are not the same
- DS is a software built upon CN
- Middleware in particular, is the software layer over the Operating system that guarantees coherence and transparency to the DS
- Example: World Wide Web is a DS built over the Internet



The Internet: a “nuts and bolts” view



Billions of connected computing **devices**:

- **hosts** = end systems
- running **network apps** at Internet's “edge”



Packet switches: forward packets (chunks of data)

- routers, switches

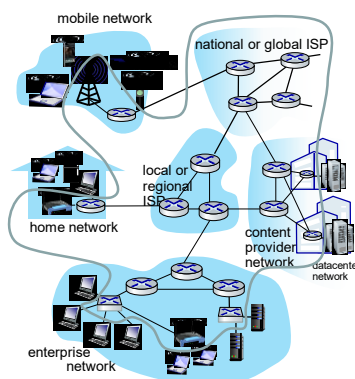


Communication links

- fiber, copper, radio, satellite
- transmission rate: **bandwidth**

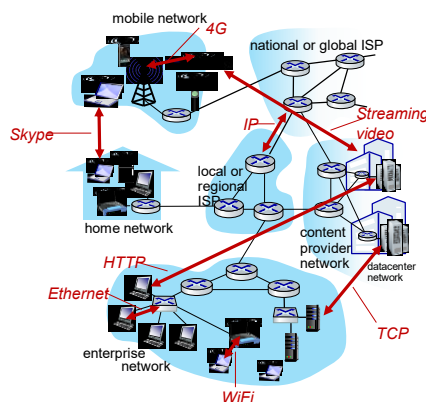
Networks

- collection of devices, routers, links: managed by an organization



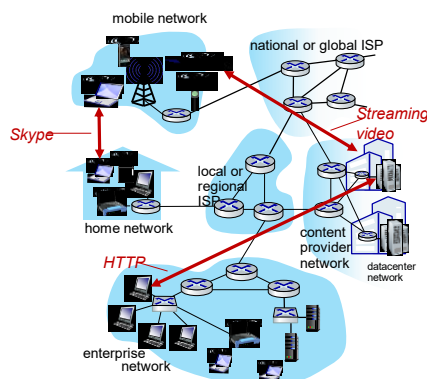
The Internet: a “nuts and bolts” view

- **Internet**: “network of networks”
 - Interconnected ISPs
- **protocols** are everywhere
 - control sending, receiving of messages
 - e.g., HTTP (Web), streaming video, Skype, TCP, IP, WiFi, 4/5G, Ethernet
- **Internet standards**
 - RFC: Request for Comments
 - IETF: Internet Engineering Task Force



The Internet: a “services” view

- **Infrastructure** that provides services to applications:
 - Web, streaming video, multimedia teleconferencing, email, games, e-commerce, social media, inter-connected appliances, ...
- provides **programming interface** to distributed applications:
 - “hooks” allowing sending/receiving apps to “connect” to, use Internet transport service
 - provides service options, analogous to postal service



Communication Systems

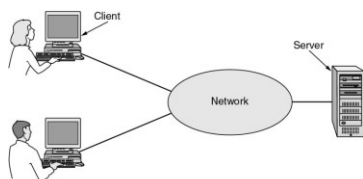
- A computer network is a communication system
- The operation of a communication systems depends on four basic characteristics:
 - Delivery
 - Precision
 - Promptness
 - Jitter

Communication System Components

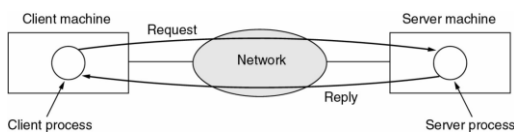
- Communication system components are:
 - Message
 - Sender
 - Receiver
 - Transmission medium
 - Protocol
- Information Data flows can be:
 - Simplex
 - Half-Duplex
 - Full-Duplex

Applications of Networks

- Why: communication and resource sharing (hardware and information)
- The client-server model is a basic paradigm model in computer networks



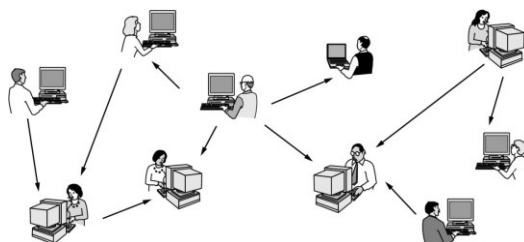
A network with two clients and one server.



The client-server model involves requests and replies

Network Applications (cont'd)

- In peer-to-peer system there are no fixed clients and servers
- This is a communication model different from the client-server model



Network Applications

- Access to remote information
- Person-to-person communication
- Interactive entertainment
- Electronic commerce
- E-learning

Network Applications (cont'd)

- Some forms of e-commerce.

Tag	Full name	Example
B2C	Business-to-consumer	Ordering books on-line
B2B	Business-to-business	Car manufacturer ordering tires from supplier
G2C	Government-to-consumer	Government distributing tax forms electronically
C2C	Consumer-to-consumer	Auctioning second-hand products on-line
P2P	Peer-to-peer	File sharing

Context for Grids, Clouds and Virtualization

- Grids promised to deliver computing power on demand combining computers from multiple administrative domains to solve a single task.
- Cloud is a recent business-oriented development for delivery of computing as service rather than as product.
- Virtualization is one of the key technologies enabling many different Clouds.

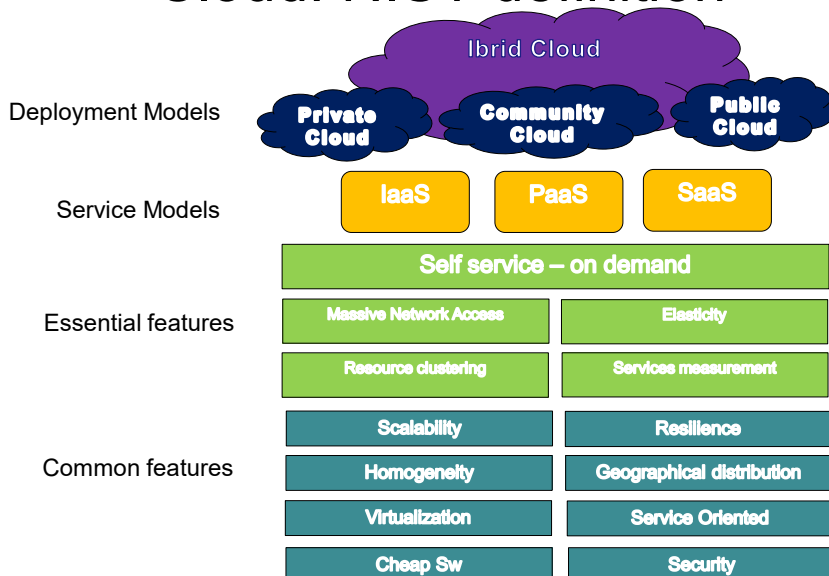
Grids

- The earliest definition of grids is focused around on-demand access to computing, data, and services
- Two years later, the definition was changed to reflect the fact that grid computing is concerned with coordinated resource sharing and problem solving in dynamic, multi-institutional virtual organizations.
- Now is intended as a grid an infrastructure for coordinated resource sharing and problem solving in dynamic, multi-institutional virtual organizations and as a system that coordinates resources that are not subject to centralized control, using standard, open, general purpose protocols and interfaces to deliver nontrivial qualities of service.

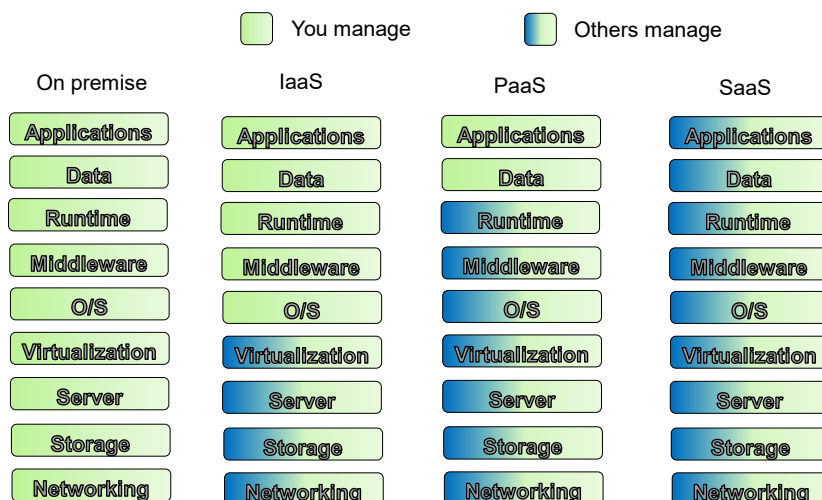
Cloud

- Three definitions for Cloud:
 - **Gartner:** Cloud computing is a style of computing where massively scalable IT related capabilities are provided “as a service” across the Internet to multiple external customers;
 - **Forrester:** A pool of abstracted, highly scalable, and managed infrastructure capable of hosting end-customer applications and billed by consumption;
 - **IBM:** An emerging computing paradigm where data and services reside in massively scalable data centers and can be ubiquitously accessed from any connected devices over the internet.

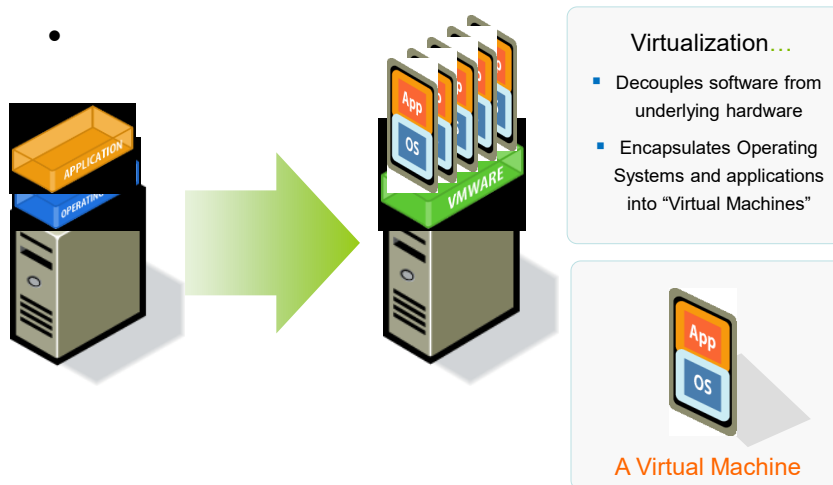
Cloud: NIST definition



Cloud: Separation of responsibility



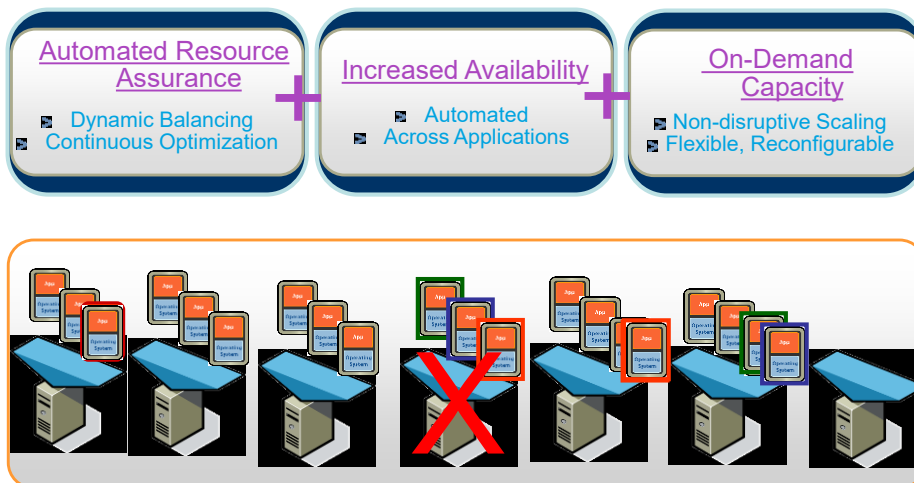
Virtualization



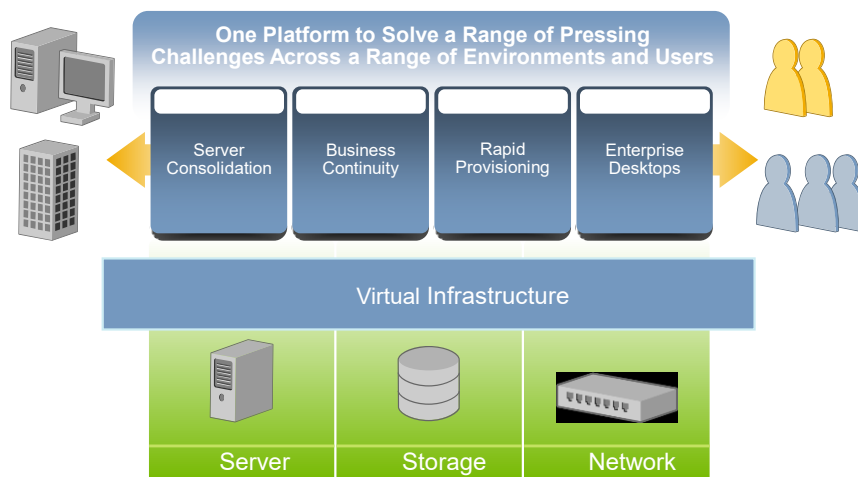
Virtualization

- A VM was an instance of the physical machine and gave users the illusion of accessing the physical machine directly.
- VMs were used to enable time-sharing and resource-sharing on the highly expensive hardware.
- Each VM was a fully protected and isolated copy of the underlying system

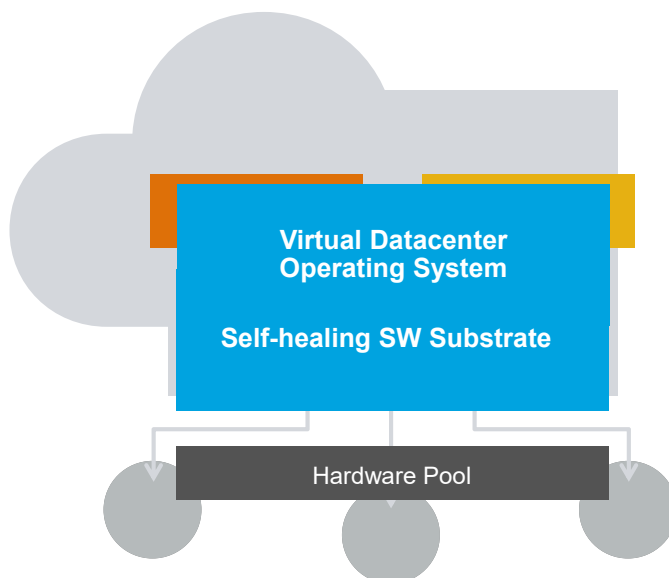
Virtual Infrastructure



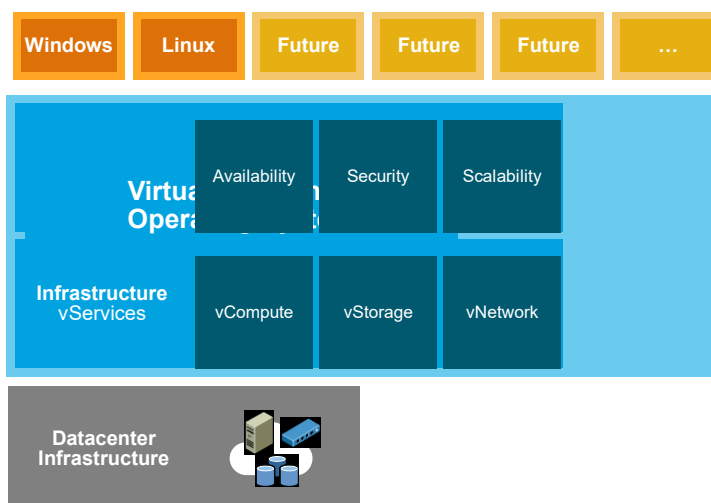
Strategic Computing Architecture



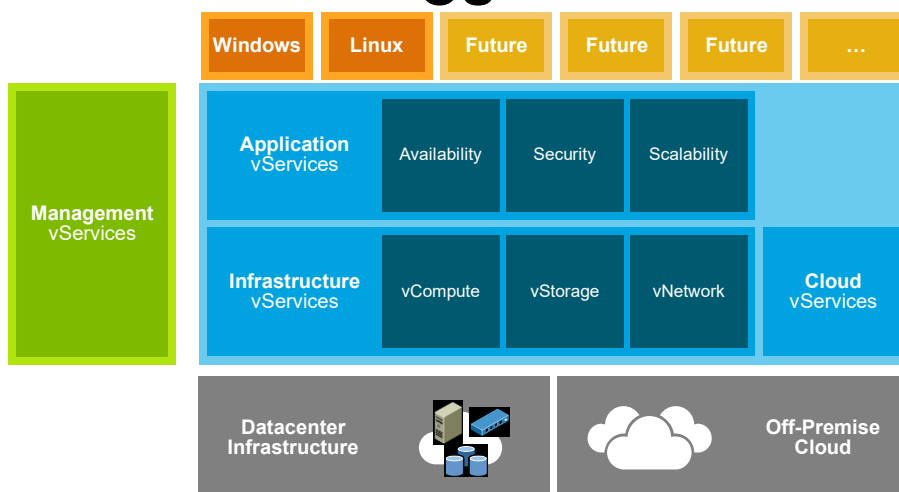
New Software Platform Required



Evolving to Virtual Data Center



Evolving to Virtual Data Center OS



Virtualization (cont'd)

- Virtualization technologies represent a **key enabler of cloud computing**.
- The most prominent feature is the **ability to install multiple OS on different virtual machines** on the same physical machine:
 - Vmware
 - Xen
 - VirtualBox
 - Parallels.

Network Classification

- Personal Area Networks
- Local Area Networks
- Metropolitan Area Networks
- Wide Area Networks
- Wireless Networks
- Home Networks
- Internetworks

Networks classification (cont'd)

- Classification of interconnected processors by scale.

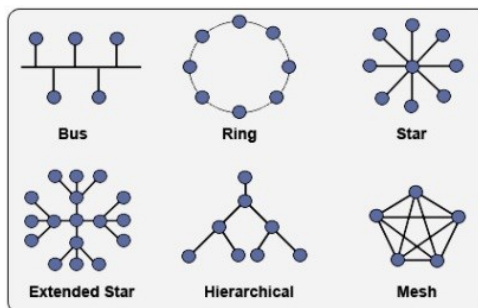
Interprocessor distance	Processors located in same	Example
1 m	Square meter	Personal area network
10 m	Room	
100 m	Building	Local area network
1 km	Campus	
10 km	City	Metropolitan area network
100 km	Country	Wide area network
1000 km	Continent	
10,000 km	Planet	The Internet

Networks classification

- Types of transmission technology
 - Broadcast links
 - Point-to-point links

Network Topologies

- Networks can have different topologies:
 - Bus
 - Ring
 - Star
 - Hierarchical
 - Mesh



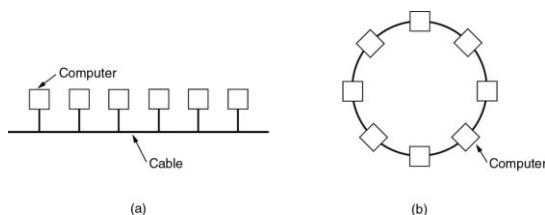
Personal Area Network

- PAN are used for the communication among the **personal devices themselves**
- Pan can be **wireless** (WiFi, Bluetooth, etc) or **wired** (USB, FireWire)



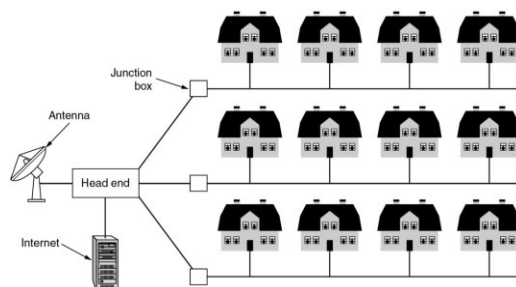
Local Area Networks

- Two broadcast networks with different topology
 - (a) Bus: 1 host can transmit in a given time instant (master) → need to **regulate conflicts** (distributed or centralized)
 - Ex: IEEE 802.3 (Ethernet, 10Mbps-10Gbps)
 - (b) Ring: bits propagate autonomously without waiting other bits in the packet. Also needs some mechanism in case of conflict
 - Ex: IEEE 802.5 (IBM token ring, 4-16Mbps)



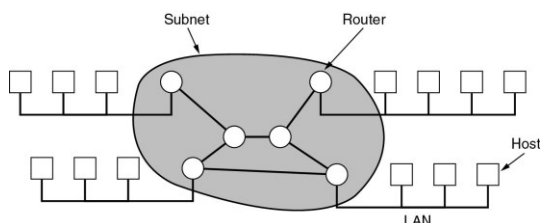
Metropolitan Area Networks

- A metropolitan area network based on cable TV.
- Other MANs: IEEE 802.16 (wireless)



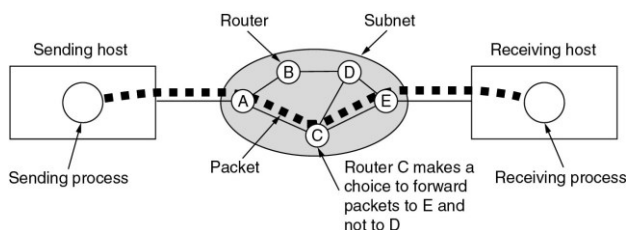
Wide Area Networks

- Relation between hosts on LANs and the subnet
 - Applications reside on hosts
 - The communication subnet is in charge of carrying packets
- Packets are stored and then forwarded by routers along the output lines
- This type of networks are called *packet-switched* or *store-and-forward* networks



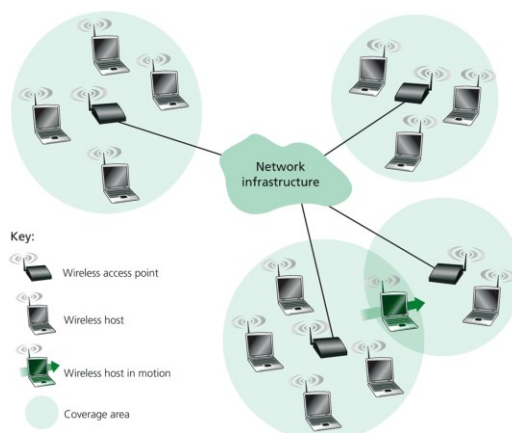
Wide Area Networks (cont'd)

- A stream of packets from **sender to receiver**
- Some networks **impose** that all packets **must follow the same path**
- In other networks each packet **can**, in principle, **follow a different path** since they are **separately forwarded**
- Paths are chosen **accordingly to routing algorithms**



A stream of packets from sender to receiver

Wireless and Mobile networks

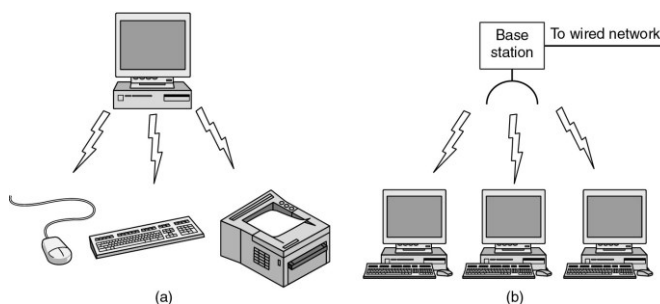


Wireless Networks

- Wireless networks can be classified in:
 - System interconnection
 - e.g. peripherals short range connections (Bluetooth)
 - Wireless LANs
 - e.g. IEEE 802.11 standard for wireless LANs
 - Wireless WANs
 - e.g. cellular phones network (150/50 Mbps for 4G)
 - e.g. IEEE 802.16 wireless MAN

Wireless Networks (cont'd)

- (a) Bluetooth configuration
- (b) Wireless LAN



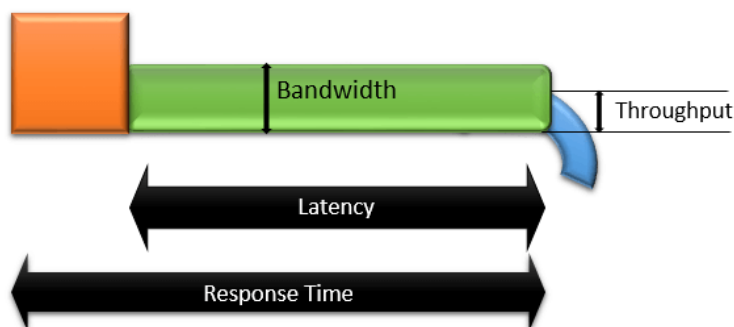
Home Network Categories

- Computers (desktop PC, PDA, shared peripherals)
- Entertainment (TV, DVD, VCR, camera, stereo, MP3)
- Telecomm (telephone, cell phone, intercom, fax)
- Appliances (microwave, fridge, clock, airconditioners)
- Telemetry (utility meter, burglar alarm, babycam).

Valutation Criteria

- Performance : delay, throughput
- Reliability: ability to delivers messages without errors
- Security: data protection.

Valutation Criteria



Protocols and Standards

- Protocols
 - Syntax
 - Semantic
 - Synchronization
- Standards
 - De facto
 - De jure

Standards Organizations

- ISO
- ANSI
- IEEE
- EIA/TIA
- CENELEC
- ETSI